

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
from google.colab import files
uploaded = files.upload()
```

[Choose files](#) train (1).csv
train (1).csv(text/csv) - 61194 bytes, last modified: 19/06/2026 - 100% done
Saving train (1).csv to train (1).csv

```
df = pd.read_csv("train (1).csv")
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	Na
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	Na
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C1
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	Na

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.isnull().sum()
```

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

dtype: int64

```
df['Age'] = df['Age'].fillna(df['Age'].mean())
```

```
df.drop(columns=['Cabin'], inplace=True)
```

```
df['AgeGroup'] = pd.cut(
    df['Age'],
    bins=[0, 12, 18, 30, 50, 80],
    labels=['Child', 'Teen', 'Young Adult', 'Adult', 'Senior']
)
```

```
df['FamilySize'] = df['SibSp'] + df['Parch']
```

```
print(df.groupby('AgeGroup')['Survived'].mean())
```

```
AgeGroup
Child      0.579710
Teen       0.428571
Young Adult 0.331096
Adult      0.423237
Senior     0.343750
Name: Survived, dtype: float64
/tmp/ipykernel_11824/3619442437.py:1: FutureWarning: The default of observed=False is deprecated
print(df.groupby('AgeGroup')['Survived'].mean())
```

```
print(df.groupby('Embarked')['Survived'].mean())
```

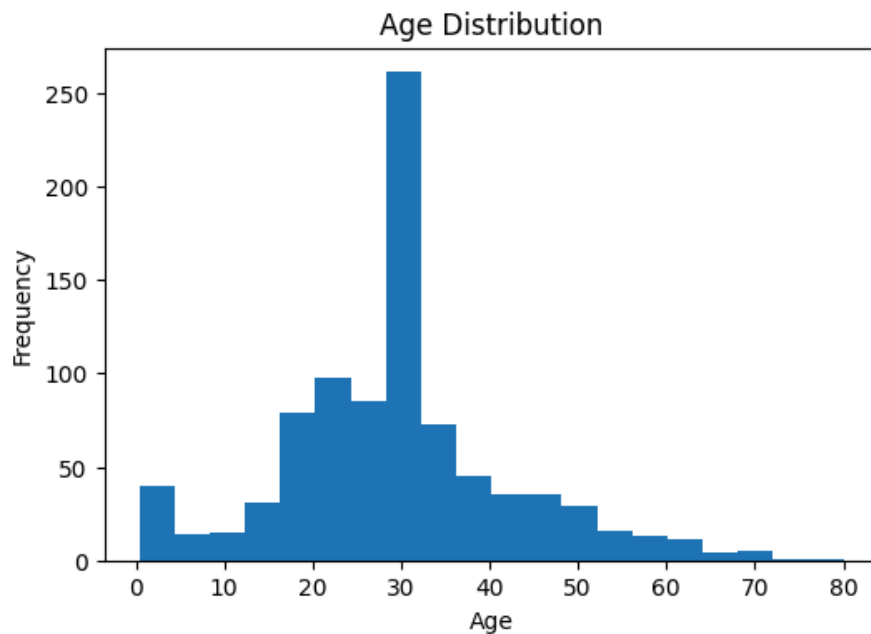
```
Embarked
C      0.553571
Q      0.389610
S      0.336957
Name: Survived, dtype: float64
```

```
print(df.groupby('FamilySize')['Survived'].mean())
```

```
FamilySize
0      0.303538
1      0.552795
```

```
2    0.578431
3    0.724138
4    0.200000
5    0.136364
6    0.333333
7    0.000000
10   0.000000
Name: Survived, dtype: float64
```

```
plt.figure(figsize=(6,4))
plt.hist(df['Age'], bins=20)
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.show()
```

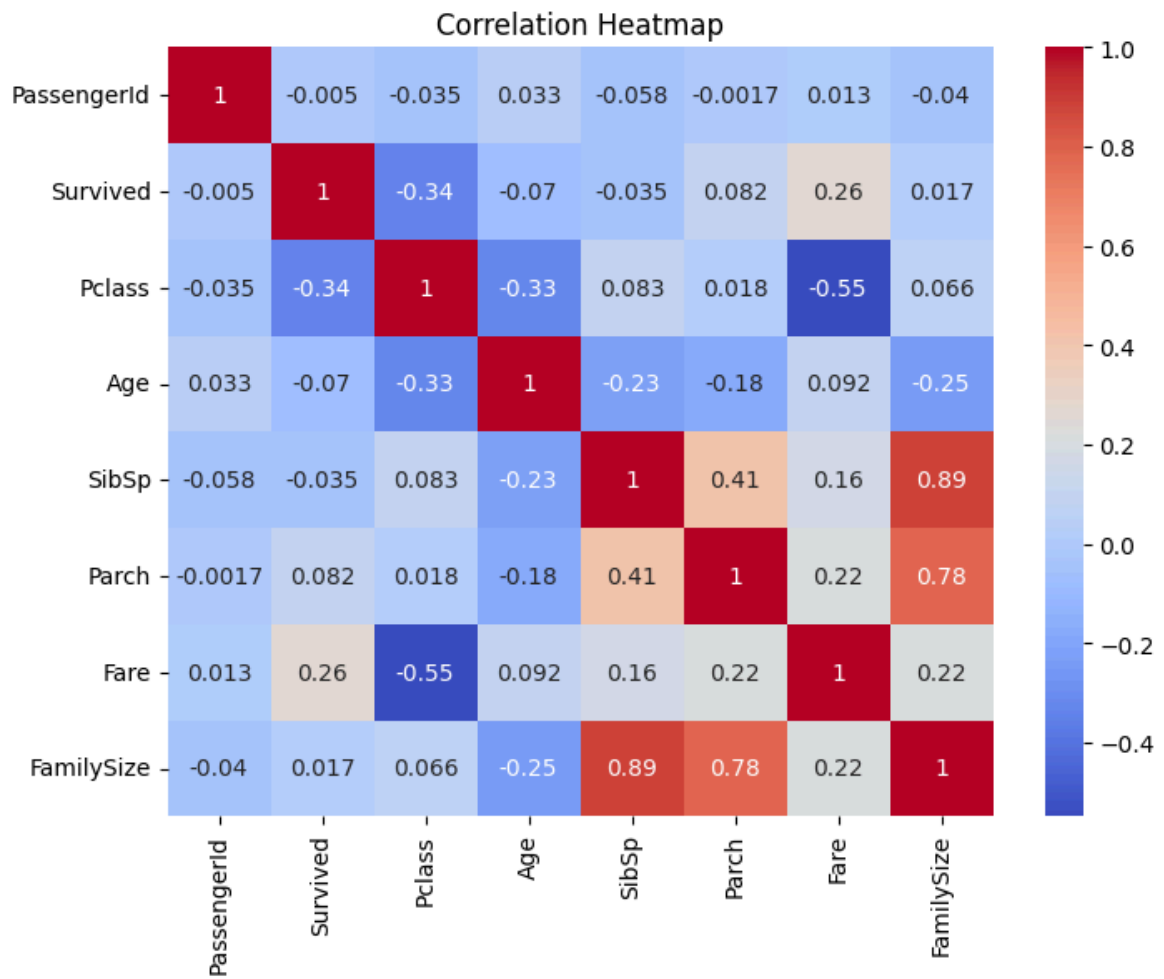


```
import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(8,6))
corr = df.select_dtypes(include='number').corr()

sns.heatmap(corr, annot=True, cmap='coolwarm')

plt.title("Correlation Heatmap")
plt.show()
```



```
df['FamilySize'] = df['SibSp'] + df['Parch']
```

```
plt.figure(figsize=(6,4))

sns.barplot(x='FamilySize', y='Survived', data=df)

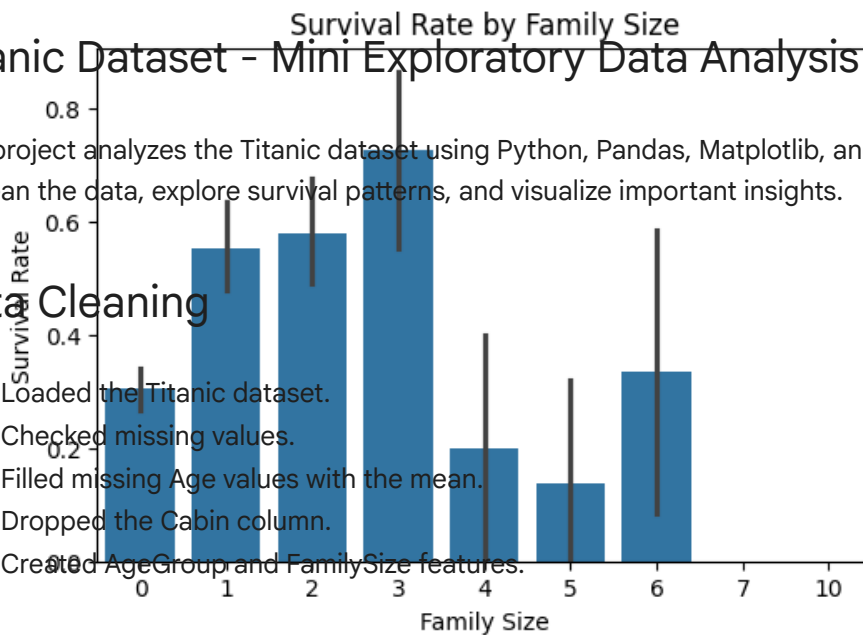
plt.title("Survival Rate by Family Size")
plt.xlabel("Family Size")
plt.ylabel("Survival Rate")
plt.show()
```

Titanic Dataset - Mini Exploratory Data Analysis (EDA)

This project analyzes the Titanic dataset using Python, Pandas, Matplotlib, and Seaborn. The objective is to clean the data, explore survival patterns, and visualize important insights.

Data Cleaning

- Loaded the Titanic dataset.
- Checked missing values.
- Filled missing Age values with the mean.
- Dropped the Cabin column.
- Created AgeGroup and FamilySize features.



Analysis

- Calculated survival rate by Age Group.
- Calculated survival rate by Embarked Port.
- Calculated survival rate by Family Size.

Visualizations

Created: